

Programming Exercises: Arrays & Matrices

A Comprehensive Collection of Algorithm Challenges

1D Array Problems

1. Print all elements of an array.
2. Find the sum of all elements in an array.
3. Find the average of array elements.
4. Find the largest element in an array.
5. Find the smallest element in an array.
6. Count even and odd numbers in an array.
7. Search for an element using linear search.
8. Reverse an array.
9. Copy one array into another array.
10. Sort an array in ascending order.
11. Sort an array in descending order.
12. Find the second largest element.
13. Find the second smallest element.
14. Count frequency of each element.
15. Find duplicate elements in an array.
16. Remove duplicate elements from an array.
17. Merge two arrays.
18. Find common elements between two arrays.
19. Check whether an array is palindrome.
20. Rotate array left by one position.
21. Rotate array right by one position.
22. Rotate array by k positions.
23. Move all zeros to the end.
24. Move all negative numbers to one side.
25. Separate even and odd elements.

26. Find missing number in an array.
27. Find pair of elements whose sum equals a target value.
28. Find all pairs with given sum.
29. Find unique elements in an array.
30. Find intersection of two arrays.
31. Find union of two arrays.
32. Find maximum difference between two elements.
33. Find minimum difference between two elements.
34. Find kth largest element.
35. Find kth smallest element.
36. Check if array is sorted.
37. Find number of occurrences of a given element.
38. Replace negative numbers with zero.
39. Calculate product of all array elements.
40. Find leader elements in an array.
41. Find equilibrium index of an array.
42. Reverse only even numbers in an array.
43. Find longest increasing subarray.
44. Find longest decreasing subarray.
45. Remove all even numbers from array.
46. Create a new array containing squares of elements.
47. Find cumulative sum array.
48. Find average of even elements only.
49. Find maximum repeated element.
50. Find minimum repeated element.
51. Print array elements in alternate order.
52. Swap first and last elements.
53. Find difference between largest and smallest element.
54. Count positive, negative, and zero elements.
55. Find prime numbers in an array.
56. Sort only even numbers in array.

57. Sort only odd numbers in array.
58. Find all subarrays of an array.
59. Find sum of all subarrays.
60. Find maximum subarray sum.

2D Matrix Problems

1. Print a 2D matrix.
2. Find sum of all elements in a matrix.
3. Find row-wise sum in matrix.
4. Find column-wise sum in matrix.
5. Find transpose of a matrix.
6. Add two matrices.
7. Multiply two matrices.
8. Find diagonal sum of matrix.
9. Find largest element in matrix.
10. Find smallest element in matrix.
11. Check whether matrix is symmetric.
12. Print boundary elements of matrix.
13. Print lower triangular matrix.
14. Print upper triangular matrix.
15. Count even and odd numbers in matrix.